

MALAVIYA NATIONAL INSTITUTE OF TECHNOLOGY JAIPUR

Department of Computer Science and Engineering
Course: Cryptography Laboratory (22CPP307) | Group: 10 (Even Group Number)

Lab Assignment 6: Cryptanalysis of Vigenère Cipher using Kasiski Examination & Frequency Analysis

Target Dataset: Ciphertext-2 (Even Group Numbers) | **Normalized Length:** 758 letters
Recovered Key: UNITEDSTATES (Length 12) | **Verification Status:** 100% Exact Match via Re-encryption

1. Implemented User-Defined Modules

- All 13 mandatory user-defined functions were implemented in Python:
- 1. clean_ciphertext():** Strips whitespaces and non-alphabetic symbols, outputting normalized uppercase letters.
 - 2. find_repeated_patterns():** Identifies repeated n-grams (lengths 3 to 5) and catalogues their exact index positions.
 - 3. calculate_distances():** Measures spacing distances between recurring pattern instances.
 - 4. find_factors():** Computes common divisors of measured distances (from 2 to 20).
 - 5. kasiski_analysis():** Ranks candidate key lengths by accumulated factor frequency counts.
 - 6. calculate_ic():** Computes Index of Coincidence ($IC = \sum f_i(f_i - 1) / (N(N-1))$) to benchmark monoalphabetic purity.
 - 7. split_into_groups():** Slices ciphertext into L independent coset streams corresponding to candidate key length.
 - 8. frequency_analysis():** Evaluates unigram letter frequencies (A-Z) per coset.
 - 9. find_shift():** Identifies Caesar shift per coset via Chi-Square goodness-of-fit against standard English frequencies.
 - 10. find_key():** Combines the optimal Caesar shifts from all cosets into the reconstructed keyword.
 - 11. vigenere_decrypt():** Decrypts ciphertext using recovered key ($P_i = (C_i - K_i) \bmod L$).
 - 12. vigenere_encrypt():** Re-encrypts recovered plaintext for verification ($C_i = (P_i + K_i) \bmod L$).
 - 13. verify():** Asserts whether re-encrypted text is character-identical to the original ciphertext.

2. Key Length Determination (Kasiski & Index of Coincidence)

Kasiski Examination: A total of 92 repeated patterns (trigrams, 4-grams, 5-grams) were identified. High-frequency factor counts clustered at 2, 3, 4, 6, and 12, indicating a base period of 12.

Index of Coincidence: Slicing into 12 cosets yielded an average IC of **0.0717**, closely converging on standard English ($IC \approx 0.068$) compared to non-period slices ($IC \approx 0.043 - 0.050$). Both tests unequivocally confirm **Key Length L = 12**.

3. Coset Analysis & Recovered Key Breakdown

Coset #	Length	Shift	Key Char	Min Chi-Square	Status
1	64	20	U	14.32	Optimal
2	64	13	N	21.86	Optimal
3	63	8	I	22.39	Optimal
4	63	19	T	34.12	Optimal
5	63	4	E	19.88	Optimal
6	63	3	D	30.49	Optimal
7	63	18	S	28.29	Optimal
8	63	19	T	22.52	Optimal
9	63	0	A	13.25	Optimal
10	63	19	T	19.90	Optimal
11	63	4	E	19.63	Optimal
12	63	18	S	24.04	Optimal

4. Plaintext Decryption & Re-Encryption Verification

Recovered Plaintext (Excerpt):

WETHEREFORETHEREPRESENTATIVESOFTHEUNITEDSTATESOFAMERICAINGENERALCONGRESSASSEMBLEDAPPEALINGTO
THESUPREMEJUDGEOFTHEWORLDFORTHERECTITUDEOFOURINTENTIONSDOINTHE NAMEANDBYAUTHORITYOFTHEGOODP
EOPLEOFTHESECOLONIESSOLEMNLYPUBLISHANDDECLARETHATTHESEUNITEDCOLONIESAREANDOFRIGHTTOUGHTTOBEFR
EEANDINDEPENDENTSTATESTHAT...

Mathematical Re-Encryption Verification:

Applying `$vigenere\_encrypt(Recovered\_Plaintext, \text{'UNITEDSTATES'})$` yielded a ciphertext matching the assigned Ciphertext-2 with **100.0% exactness across all 758 characters**.